Huawei New-Gen OceanStor A800 High-Performance AI Storage

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Data Sheet

- Ultra Performance
- Omni-Expansion
- AI Native

Huawei OceanStor A800 is the new-gen high-performance distributed storage designed for AI. Its innovative data and control plane separation architecture enables ultra performance and accelerates large AI model training and inference. OceanStor A800 offers 500 GB/s bandwidth and 10 million IOPS per controller enclosure, 4x faster training set loading, and 3x faster training resumption from checkpoints than the leading alternative.

Huawei OceanStor A800 high-performance AI storage also supports flexible scalability, with a single cluster able to scale out to 512 controllers and EB-level storage capacity expansion needed for the smooth evolution of multimodal models with trillions or tens of trillions of parameters. OceanStor A800 supports new data paradigms, like vectors, with its built-in high-performance vector retrieval engine that accelerates retrieval from large-capacity vector knowledge bases and reduces AI hallucinations. It also provides multi-level KV cache to reduce inference latency while improving cost-effectiveness. This enables a single storage system to meet the data processing needs of the entire AI training and inference process.

OceanStor A800 high-performance AI storage

Product Introduction

- OceanStor A800 is an all-flash high-performance AI storage product that delivers excellent performance, high capacity, and elastic scalability based on brand-new hardware. It integrates disks and controllers, with each controller enclosure supporting 64 high-speed NVMe SSDs and providing raw capacity ranging from 245.76 TB to 983.04 TB. With ultimate scalability, a single cluster can scale out to 512 controllers, reaching EB-level capacity.

- Based on the brand-new data and control plane separation architecture, OceanStor A800 high-performance AI storage provides up to 500 GB/s bandwidth and 10 million IOPS. A single storage system accelerates the entire AI process, from data ingestion to training and inference. OceanStor A800 is widely applicable to financial AI models, regional intelligent computing centers, AI computing power centers, scientific research platforms, autonomous driving platforms, and more.

Product Features

Ultra Performance for Higher AI Cluster Computing Power Utilization

OceanStor A800 uses NPUDirect Storage (NDS) to build an optimal performance path from NPUs to data storage, bypassing complex protocol stacks and eliminating the need for data copy, improving performance by 40%.

New-gen OceanStor A800 AI storage uses the DataTurbo high-performance file acceleration engine to directly read data from and write data to the storage system in kernel mode. Kernel-mode caching enables memory data mobility and halves memory usage. OceanStor A800 offers 500 GB/s bandwidth per controller enclosure, over 100 TB/s aggregated bandwidth per cluster, read and write for TB-level checkpoints in seconds when training models with trillions of parameters, and fast resumable training from checkpoints.

Based on a global unified data view, OceanStor A800 uses DPUs to offload the general computing power of CPUs. Data read and write operations are processed by DPUs, bypassing the CPUs and memory. The host can thus directly connect to disks to read and write data. A single controller enclosure provides 10 million IOPS and a cluster provides hundreds of millions of IOPS, with four times higher data loading efficiency than the leading alternative.

More Flexible Multi-Dimensional Expansion

Based on the brand-new hardware architecture, new-gen OceanStor A800 AI storage delivers flexible scalability with both scale-out and scale-up capabilities. A single cluster can scale out to up to 512 controllers and EB-level storage capacity, ensuring high performance and capacity density. In addition, computing cards, such as DPU and GPU accelerator cards*, can be inserted into the back end of the array. The storage cluster can scale up to a maximum of 4,096 cards, together to provide 14,400 TFLOPS of computing power and accelerate data processing. It can also carry multiple offloaded functions, including encryption, compression, and vector retrieval.

A single new-gen OceanStor A800 AI storage system can meet diverse ecosystem requirements. It is compatible with file, object, and vector, among other protocols, and supports the entire large AI model development process, such as data ingestion, preprocessing, model training, and deployment and inference. OceanStor A800 eliminates the need for data copy and accelerates the iteration and development of large AI models.

New-gen OceanStor A800 AI storage provides SmartQuota. With this file system quota technology, system administrators can manage and control the storage resources. By limiting the disk space that can be used by specified subjects, SmartQuota prevents excessive resource occupation and maximizes the value of storage space.

More Efficient Inference with AI-Native Data Paradigms

The data paradigms supported by new-gen OceanStor A800 AI storage range from NAS, and objects to new vectors and graph adaptive retrieval engines. OceanStor A800 enables high-performance retrieval from massive amounts of multimodal knowledge. The solution's competitive advantage lies in its inference based on RAG knowledge bases and Unified Cache. OceanStor A800 provides a built-in large-capacity AI knowledge base with hundreds of millions of data records for 3x higher QPS than the leading alternative. This enables faster knowledge base creation and more accurate queries.

OceanStor A800 significantly accelerates inference thanks to its Unified Cache feature, which offers multi-level KV cache from on-chip memory to SSDs. This establishes a PB-level global shared resource pool for KV cache, enabling flexible allocation across multiple compute nodes and xPUs and eliminating repeated computing. This slashes the Time to First Token (TTFT) by 78% while improving the throughput by 60%. It also uses sparse attention algorithms for retrieval to compress tokens to enable the inference of ultra-long sequences beyond the context window, ultimately making inference smoother, faster, and more cost-effective.

Key Specifications

Model	OceanStor A800
Hardware Architecture	Disk and controller integration
Maximum Raw Capacity per Controller Enclosure	983.04 TB
Height of Each Controller Enclosure	8 U
Controllers per Controller Enclosure	2
Disks per Controller Enclosure	64
Processors per Controller	Four Kunpeng 920 processors
Maximum Memory per Controller	1024 GB
Data Disk Type	Palm-sized NVMe SSD
Network Type	• 25/100 Gbit/s TCP/IP • 25/100/200 Gbit/s RoCE • 200 Gbit/s InfiniBand
Key Feature	Cluster (HyperScale), quota (SmartQuota), Quality of Service (SmartQoS), multi-protocol interworking (SmartInterworking)*, end-to-end data integrity check (DIF)
Enclosure Dimensions (H x W x D)	352.8 mm × 447 mm × 952.2 mm
Operating Temperature	–60 m to +1800 m altitude: 5°C to 35°C 1800 m to 3000 m altitude: The max. temperature threshold decreases by 1°C for every altitude increase of 220 m
Operating Humidity	10% to 90% RH

* If this feature is required, contact the sales personnel.

To learn more about Huawei storage, please contact your local Huawei office or visit the Huawei Enterprise website: https://e.huawei.com.

Huawei IT

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